

American Sign Language Recognition with Convolutional Neural Networks: A Gateway to Enhanced Inclusivity

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Abstract. In the realm of higher education, where the right to inclusive education is fundamental, it is essential to address the distinctive challenges faced by students with disabilities, particularly those who are deaf. This study explores inadequacies in accommodating the needs of deaf students within a university setting through the implementation of a Convolutional Neural Network (CNN) for sign language recognition. We conducted thorough testing of our model using three optimizers Adaptive Moment Estimation (ADAM), Stochastic Gradient Descent (SGD), and Stochastic Gradient Descent (SGD) with Nesterov's Momentum. Notably, the highest test accuracy of 96% was attained when employing SGD with Nesterov momentum on the Sign MNIST dataset. These findings underscore that our proposed approach outperforms other state-of-the-art methods.

Keywords: Inclusive Education · Sign Language Recognition · Convolutional Neural Network (CNN) · Deep Learning

1 Introduction

In the continually advancing realm of inclusive education, there is a growing recognition of the need to address the unique requirements of students with disabilities, particularly those who are deaf or hard of hearing and rely on sign language as their primary mode of communication. Traditional educational methods often struggle to fully accommodate these students' communication needs, presenting barriers to effective learning and participation. Recent research has responded to this challenge by harnessing the progress in artificial intelligence, machine learning, and deep learning to offer tailored support for students with disabilities. Inclusive classrooms emphasize the importance of diverse communication modes such as sign language, communication aids, or augmented and alternative communication methods [1]. Nevertheless, many educators encounter difficulties in managing this diversity due to insufficient training and support [2]. Information and Communication Technology (ICT) emerges as a pivotal tool in fostering e-inclusion,

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necessitating accessible resources and teacher competencies for genuine inclusive education [2]. Highlighting the significance of adaptable teaching materials and localized approaches, Huang et al. [1] underscore the role of ICT in enabling successful education, particularly in marginalized settings like rural areas. Echoing these sentiments, Drigas and Ioannidou [3] emphasize how ICT can revolutionize special education, enabling comprehensive assessment and effective interventions to enrich the learning experiences of students with special educational needs.

Recognizing these challenges, this article embarks on an exploration of an innovative solution that seeks to revolutionize the educational landscape for individuals reliant on sign language. At the heart of this transformative approach is the integration of Convolutional Neural Network (CNN) [4] technology. The choice of CNNs is strategic, as these neural networks excel in pattern recognition tasks, making them particularly well-suited for the intricate and nuanced nature of sign language interpretation.

The use of sign language as a primary means of communication stands out as an important feature within the deaf and hard-of-hearing community. This underscores the importance of developing tailored tools and approaches to cater to their unique communication needs. While conventional educational methods may struggle to adapt to these unique requirements, the infusion of CNN technology aims to bridge the communication gap. By leveraging the power of deep learning and neural networks, we aspire to create a more inclusive and effective educational environment for individuals with hearing and speech impairments.

The main objectives of this study can be outlined as follows:

- Introducing an innovative sign language model utilizing Convolutional Neural Network CNN.
- Assessing the effectiveness of the model in recognizing sign language through assessments using different pre-trained model optimizers on the Sign MNIST dataset.

The subsequent sections of this article are organized as follows: Sect. 2 presents related work, while Sect. 3 introduces the proposed methodology in the domain of sign language recognition. Section 4 outlines our proposed CNN-based methodology and discusses the obtained results. Finally, Sect. 5 encapsulates the conclusion.

2 Related Work

In the realm of inclusive education and assistive technology, several studies have investigated the application of artificial intelligence (AI) and deep learning algorithms to facilitate communication and learning for individuals with disabilities. Srivastava [5] emphasizes the importance of AI-driven Smart Learning Assistance tools in fostering inclusive education, ensuring equitable access, and efficient communication for students with disabilities in educational settings. Additionally, Jarbou et al. [6] demonstrate the efficacy of deep learning algorithms in forecasting school absenteeism among autistic students, potentially amplifying their access to crucial educational and therapeutic resources while mitigating the likelihood of dropout.

Jain et al. [7] utilized CNNs to recognize American Sign Language and highlighted the significance of filter size, demonstrating the potential for improvements. Mittal et al.

[8] designed a sign language recognition system using LSTM and Leap Motion. They applied a four-gated LSTM cell with a 2D CNN for sentence recognition, achieving 72.3% accuracy. Individual word recognition attained an accuracy of 89.50% with this system. Aparna and Geetha [9] propose a pioneering sign language recognition method, blending CNNs and LSTM networks. They use pre-trained CNNs for feature extraction, followed by LSTM layers, achieving a notable 94% accuracy on the Indian Sign Language (ISL) dataset, demonstrating its significance in deep learning-based sign language recognition.

3 Proposed Methodology

3.1 American Sign Language Recognition Using Convolutional Neural Network (CNN)

In our research, a customized convolutional neural network (CNN) [4] model tailored for image recognition was utilized. The first convolutional layer applies 128 filters, succeeded by max pooling for dimension reduction. A second convolutional layer with 64 filters is introduced, followed by another max-pooling layer. Subsequently, a third convolutional layer with 32 filters is implemented, along with a corresponding max-pooling layer. The output is then flattened into a one-dimensional array before traversing through a fully connected layer with 7 neurons and a ReLU activation. A dropout layer is strategically incorporated to mitigate overfitting, deactivating 25% of neurons. The final layer comprises 24 neurons with Softmax activation for multi-class classification. This comprehensive architecture aligns with our overarching goal of advancing sign language recognition technology and promoting inclusivity within educational environments. Figure 1 illustrates a flowchart of our proposed methodology, while Table 1 presents the model summary of the CNN model.

3.2 Models Optimization

In order to optimize the categorical cross-entropy loss function of our models, defined in the following equation:

$$j(\theta) = \frac{-1}{m} \sum_{i=1}^{m} \sum_{j=1}^{c} y_{ij} \log(p_{ij}) \tag{1}$$

where m is the number of samples, C is the number of classes, Y_{ij} is an indicator of whether class j is the correct classification for sample i, and p_{ij} is the predicted probability of sample i belonging to class j, we employed a range of optimization algorithms. This range included Stochastic Gradient Descent (SGD) [10], Adaptive Moment Estimation (ADAM) [11], and SGD with Nesterov Momentum [12]. This diverse selection was made with the specific goal of thoroughly assessing and comparing the impact of different optimization methods on the performance in our study.

- **Stochastic Gradient Descent (SGD)** [10]**:** is a widely utilized optimization algorithm in the field of machine learning, particularly for training complex models like neural networks. Its fundamental principle revolves around iteratively updating model

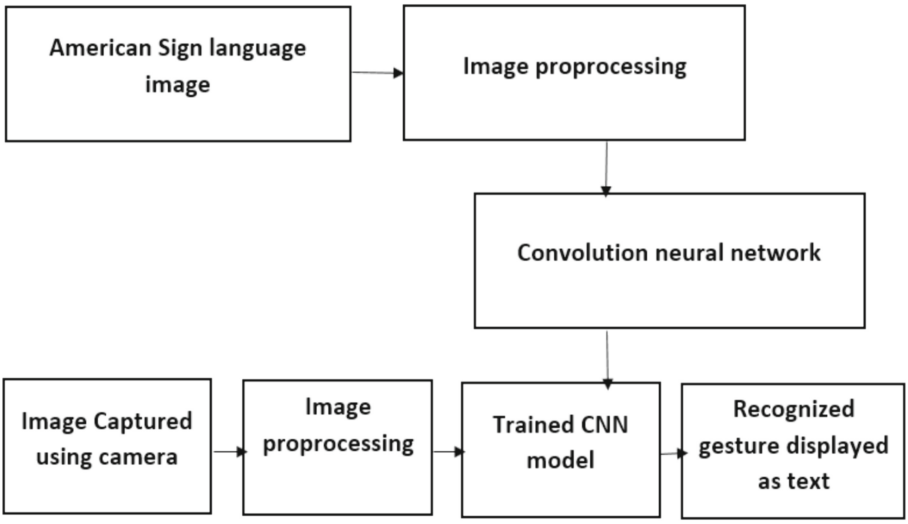


Fig. 1. Flowchart of American sign language recognition system using convolutional neural networks

Table 1. Model summary of the CNN

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 28, 28, 128)	3328
max_pooling2d (MaxPooling2D)	(None, 14, 14, 128)	0
conv2d_1 (Conv2D)	(None, 14, 14, 64)	32832
max_pooling2d_1 (MaxPooling2D)	(None, 7, 7, 64)	0
conv2d_2 (Conv2D)	(None, 7, 7, 32)	8224
max_pooling2d_2 (MaxPooling2D)	(None, 4, 4, 32)	0
flatten (Flatten)	(None, 512)	0
dense (Dense)	(None, 7)	3591
dropout (Dropout)	(None, 7)	0
dense_1 (Dense)	(None, 24)	192

Total params: 48167 (188.15 KB).
Trainable params: 48167 (188.15 KB).
Non-trainable params: 0 (0.00 Byte).

parameters using a stochastic estimate of the gradient of the loss function concerning these parameters.

- Consider a parametric model $f(x;\theta)$ where x is the input, θ represents the model parameters, and $f(x;\theta)$ is the model prediction. The objective is to minimize a loss function $J(\theta)$ associated with the difference between the model predictions and true values. The parameter update with SGD can be mathematically formulated as follows:

$$\theta_{t+1} = \theta_t - \eta \nabla_\theta J_B(\theta_t) \tag{2}$$

where θt is the parameter vector at iteration t, η is the learning rate, a hyperparameter controlling the size of the update steps, JB(θt) is the loss function, and ∇θJB(θt) is the gradient of the loss function with respect to parameters θt.

- **Stochastic Gradient Descent (SGD) with Nesterov's Momentum** [12]**:** is an optimization algorithm employed in machine learning, especially for training neural networks. It enhances the standard SGD algorithm by incorporating Nesterov's Momentum, a modification of traditional momentum. Nesterov's Momentum introduces a look-ahead mechanism, anticipating the next parameter update and adjusting the optimization trajectory accordingly. Mathematically, the update rules for Nesterov's Momentum within the SGD framework can be expressed as follows:

$$\begin{aligned} v_t &= \gamma v_{t-1} + \eta \nabla J(\theta - \gamma v_{t-1}) \\ \theta_{t+1} &= \theta_t - v_t \end{aligned} \tag{3}$$

where, v_t represents the velocity term, γ is the momentum parameter, v_{t-1} is the velocity at the previous time step, η is the learning rate, ∇J is the gradient of the loss function, and θ_t denotes the current parameters. The parameter update is performed using the velocity term v_t, contributing to improved convergence speed and stability, particularly in scenarios with complex optimization landscapes.

- **Adaptive Moment Estimation (ADAM)** [11]**:** is a commonly used method. The moment update is converted into an exponential moving average. The equation of the ADAM optimization algorithm is represented as follows:

$$\theta_{t+1} = \theta_t - \alpha \sqrt{v_t} + \varepsilon \cdot m_t \tag{4}$$

where θ_{t+1} represents the updated parameters at time step $t + 1$, θ_t represents the current parameters at time step t, α denotes the learning rate, v_t represents the average of squared gradients diminishing exponentially, ϵ is a small constant (usually around 1×10^{-8}) for numerical stability, and mt represents the exponentially decaying average of gradients.

3.3 Models Evaluation

- **Accuracy:** defined in (5), represents the ratio of the number of correct predictions to the total number of observations in the dataset

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{5}$$

- **Precision:** defined in (6), is a metric assessing the accuracy of a model in recognizing pertinent objects. Calculated as the ratio of True Positives (TPs) to the overall detections made by the model, precision essentially measures the model's capability to accurately pinpoint and categorize objects of interest within its predictions.

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} \ + \ \text{False Positives}} \tag{6}$$

- **Recall:** defined in (7), assesses the model's ability to recognize all true instances, indicating the ratio of True Positives (TPs) to the total actual instances. A well-suited model is characterized by both high precision and high recall. A flawless model, where Precision equals 1 and Recall equals 1, implies the complete elimination of False Negatives (FNs) and False Positives (FPs). However, achieving such perfection is often a challenging task.

$$Recall = \frac{True\ Positives}{True\ Positives + False\ Negatives} \tag{7}$$

- **F1-score:** defined in (8), measures the harmonic mean between recall and precision.

$$F1 = 2 \times \frac{(Precision \times Recall)}{(Precision + Recall)} \tag{8}$$

4 Experimental Results

4.1 Setup Environment

The experiment was conducted using a system running Windows 10 with 16 GB of RAM, a 64-bit architecture, a CPU with a clock speed of 3.20 GHz, Nvidia GTX 1070 graphics card, and an Intel Core i7 processor. We employed the Python-based Keras library [13] to facilitate the integration of models featuring different layer configurations into the training procedure. Before being fed into the system, images underwent resizing to dimensions of 224 $\times$ 224 pixels. Our model of choice was the Convolutional Neural Network (CNN). The entire process was conducted within the Google Colab platform, utilizing GPUs to ensure consistent conditions across all four models. Throughout the training phases, vigilant observation of the model weights enabled us to pinpoint the stage of peak validation accuracy.

4.2 Sign MNIST Dataset

The dataset used in this study has been carefully designed to closely resemble the well-known sign MNIST dataset [14], as shown in Fig. 2, providing a familiar framework for researchers and practitioners. Essentially, each training and test case, aligning with the inherent challenge of accurately capturing gesture nuances.

Regarding size, the training data consists of 27455 individual cases, each uniquely labeled and prepared for analysis. Similarly, the test dataset comprises 7172 cases meticulously curated for validation purposes. While slightly smaller than the traditional MNIST handwritten digit dataset, it maintains a comparable underlying structure, facilitating seamless comparisons and analyses. The dataset follows a consistent format, starting with a header row containing essential information such as the label associated with a particular case. Subsequent columns represent pixel values, numbered from pixel1 to pixel784. Each pixel value indicates the grayscale intensity of a single pixel in a 28x28 image, ranging from 0 to 255. This detailed pixel-level representation serves as the foundation for capturing the visual intricacies of the images in the dataset, supporting subsequent analysis and recognition efforts.



Fig. 2. Example of images from the sign language MNIST dataset

4.3 Results

Figure 3 presents the performance of a Convolutional Neural Network (CNN) model in terms of various metrics. The 96% accuracy indicates that the model correctly classified 96% of the data. With a precision of 99%, the model is accurate 99% of the time when predicting the positive class. A recall of 98% means that the model correctly identified 98% of instances belonging to the positive class. Lastly, the 99% F-score represents a harmonic mean between precision and recall, indicating a high overall balance between these two measures. These results suggest that the CNN model demonstrates strong and balanced performance across the metrics, although evaluation should always be interpreted in the context of the specific nature of the problem and dataset characteristics.

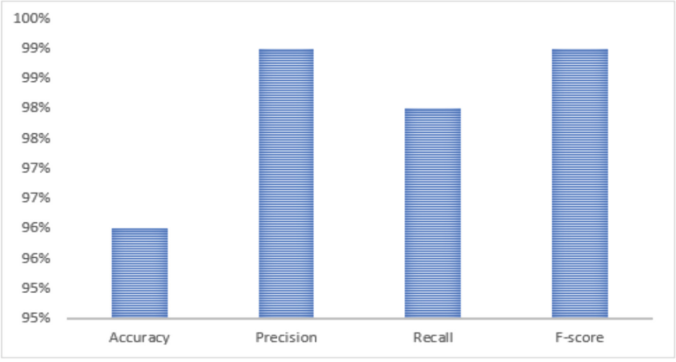


Fig. 3. Performance metrics for CNN-based sign language model

In the evaluation of different CNN models, three optimization algorithms were employed, as shown in Fig. 4. Stochastic Gradient Descent (SGD) with Nesterov momentum achieved the highest accuracy at 96%, showcasing its effectiveness in converging to

optimal solutions during training. The second model, utilizing plain Stochastic Gradient Descent (SGD), achieved a commendable accuracy of 94.70%, demonstrating the robustness of traditional SGD optimization. Lastly, the ADAM optimizer yielded an accuracy of 90%, indicating its slightly lower performance compared to the SGD variants in the specific context of the CNN models under consideration.

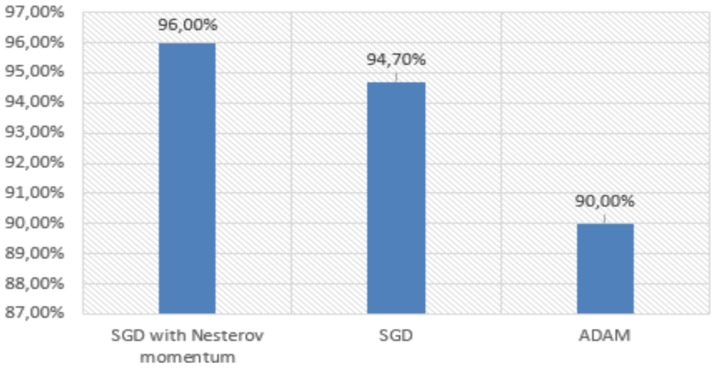


Fig. 4. Performance comparison of optimizers based on test accuracy

Table 2 provides a comparative overview of Sign Language Recognition Accuracy across different methodologies. The first approach, employing 2D-CNN and Modified LSTM with a Leap motion sensor [8], achieves an accuracy of 89.50% on a custom dataset with 3150 signs. In contrast, a CNN and 2-layer LSTM [9] model attain a higher accuracy of 94% on a smaller custom dataset of 6 signs. The third method, utilizing CNN [15], achieves an accuracy of 94.7% on the NUS hand Posture dataset and American Fingerspelling A Dataset. Notably, the proposed approach in the study outperforms all others, achieving a remarkable 96% accuracy, leveraging the MNIST dataset for evaluation.

Table 2. Comparison of sign language recognition accuracy among different approaches

Methodology	Dataset	Accuracy
2D-CNN and Modified LSTM with Leap motion sensor [8]	Custom Dataset (3150 signs)	89.50%
CNN and 2layer LSTM [9]	Custom Dataset (20 videos)	94%
CNN [15]	NUS hand Posture dataset, American Dataset	94.7%
Our proposed approach	MNIST dataset (American)	96%

5 Conclusion

In conclusion, our research underscores the pivotal role played by Convolutional Neural Networks (CNNs) in advancing the domain of sign language recognition, especially within the realms of higher education and inclusive learning environments. The robust methodology we proposed, harnessing a meticulously curated dataset and customized optimization techniques, has yielded remarkable success, boasting a test accuracy of 96%. This surpasses the performance of other state-of-the-art methodologies. The precision, recall, and F-score of our model further accentuate its proficiency in accurately classifying sign language gestures, an important aspect for effective communication in educational settings. These results not only highlight the viability of our CNN-based approach but also underscore its potential to foster more inclusive educational environments by facilitating communication for students who are deaf or hard of hearing. The adaptability of our model, striking a balance between precision and recall, ensures reliable and consistent performance across various metrics. Looking ahead, our work contributes significantly to the ongoing discourse on leveraging artificial intelligence to enhance accessibility and inclusivity. The success of our methodology serves as a solid foundation for continued exploration and development, encouraging collaboration among academia, technology developers, and educational institutions to create a more equitable and accessible learning experience for all students, irrespective of their abilities or disabilities. Through ongoing research and implementation efforts, we envision a future where advanced technologies like Deep CNN play a pivotal role in breaking down communication barriers and promoting inclusivity in educational settings globally.

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